bytes from one socket to another. In the IoT world, things are more complicated for many reasons:

- The IoT device may be *constrained*, reducing its ability to execute the public key cryptography used in TLS. For example, the device may have limited CPU power and memory, or may be operating purely on battery power that it needs to conserve.
- For efficiency, devices often use compact binary formats and low-level networking based on UDP rather than high-level TCP-based protocols such as HTTP and TLS.
- A variety of protocols may be used to transmit a single message from a device to its destination, from short-range wireless protocols such as Bluetooth Low Energy (BLE) or Zigbee, to messaging protocols like MQTT or XMPP. Gateway devices can translate messages from one protocol to another, as shown in figure 12.1, but need to decrypt the protocol messages to do so. This prevents a simple end-to-end TLS connection being used.
- Some commonly used cryptographic algorithms are difficult to implement securely or efficiently on devices due to hardware constraints or new threats from physical attackers that are less applicable to server-side APIs.

DEFINITION A ***constrained device*** has significantly reduced CPU power, memory, connectivity, or energy availability compared to a server or traditional API client machine. For example, the memory available to a device may be measured in kilobytes compared to